

AWS Assignment

Course Name & Code: AWS

Instructor: Nitha

Assignment Details

Assignment Title: Task #48793 - Backup Scripting with S3 Synchronization

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Instructions and Guidelines

Objective

GOAL: send an email to the verified address using aws-SES with php

1. verify the recipient address inside a simple email service[SES].
2. Get your smtp credentials (from SES console)
3. install php and other required dependencies in your machine
4. try to send mail by creating and executing a php file [refer aws docs for the information about the file and its contents]

Result

Step 1: IAM Policy and Role Creation

A custom IAM policy named **S3BackupPolicy** was created to provide the necessary S3 permissions. This policy was then attached to a new IAM role, **EC2-S3-Backup-Role**, which was assigned to the EC2 instance to allow secure communication with S3 without using static credentials.



Figure 1: S3BackupPolicy details.

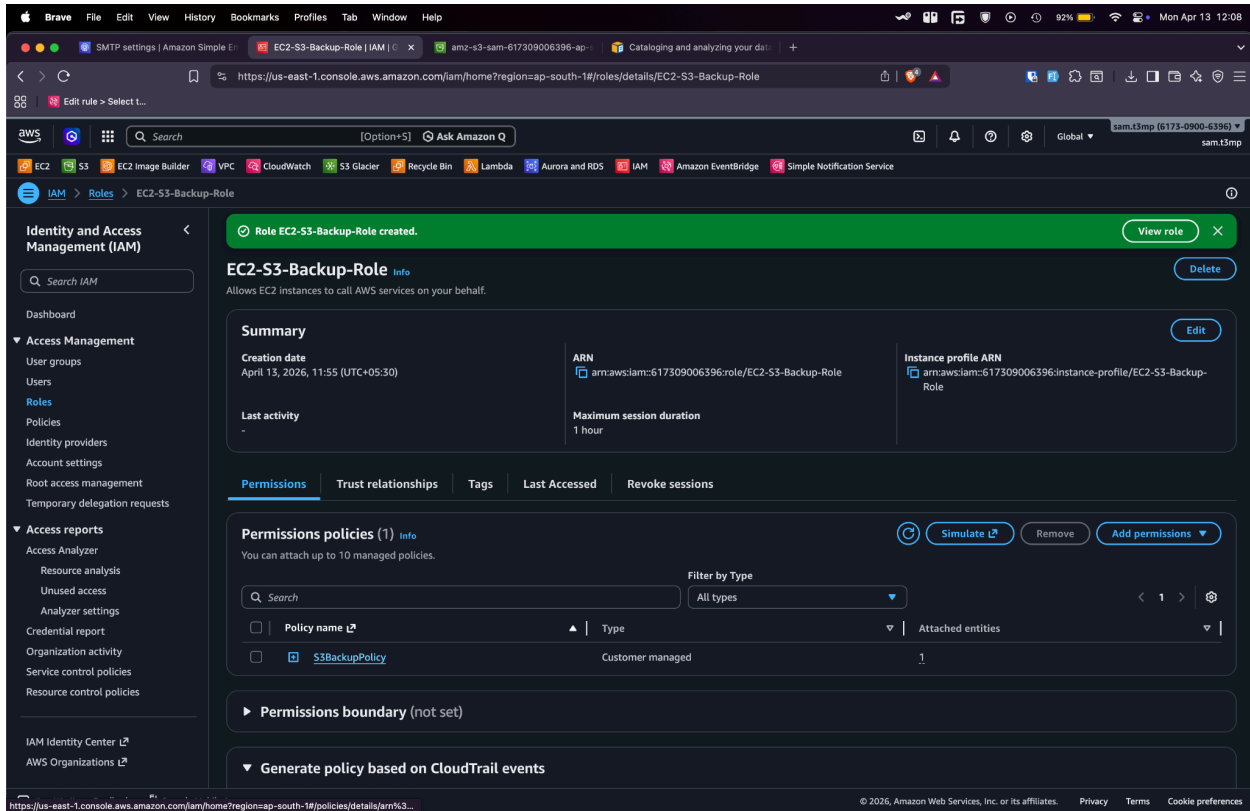


Figure 2: EC2-S3-Backup-Role attached to the instance.

Step 2: Database and Script Configuration

MariaDB was configured on the server, and a shell script (backup_script.sh) was programmed using **Vim**. The script automates the full database dump and utilizes the aws s3 sync command to transfer files to the target bucket: amz-s3-sam-617309006396-ap-south-1-an.

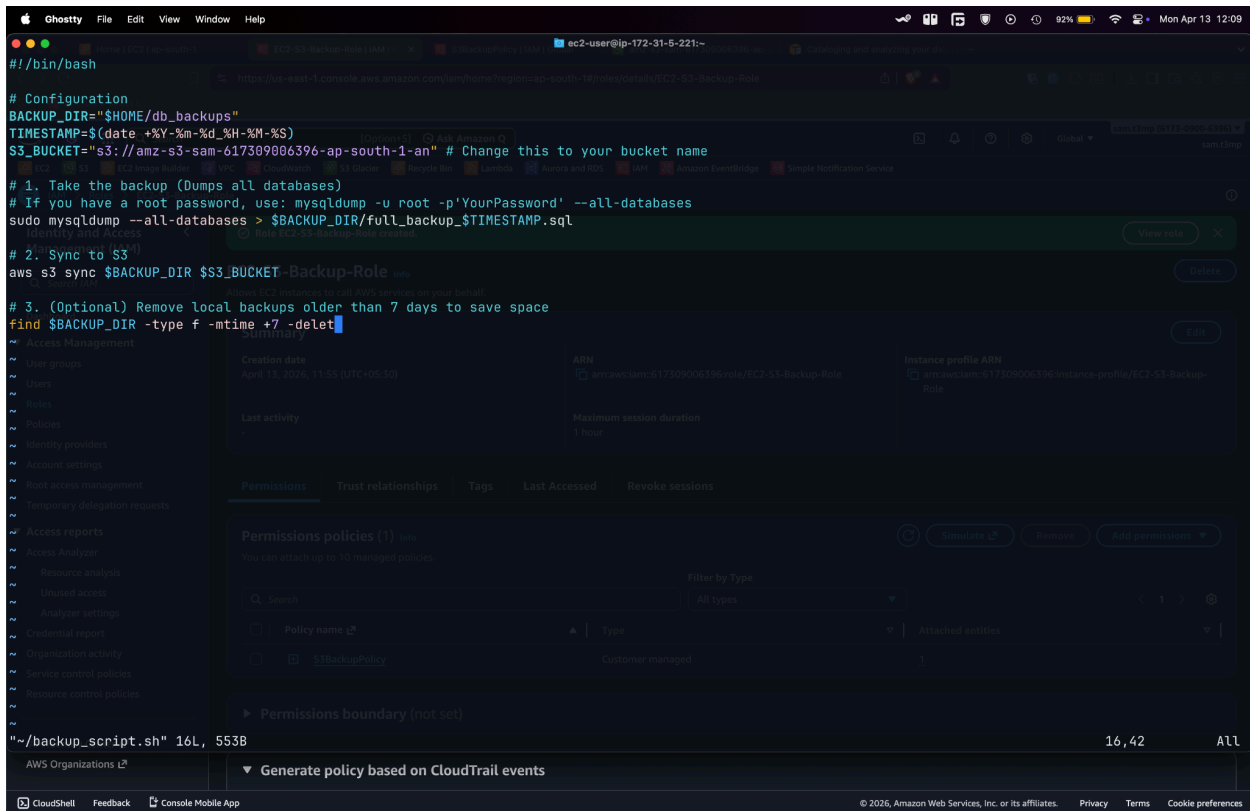


Figure 3: Bash script logic for automated backups.

Step 3: Execution and Synchronization

The AWS CLI was configured with the appropriate region (ap-south-1). Upon execution, the script successfully generated the backup file and synchronized it to the S3 bucket as shown in the terminal logs.

```

Installing      : python3-prompt-toolkit-3.0.41-5.el10.noarch
Installing      : python3-ply-3.11-25.el10.noarch
Installing      : python3-pycparser-2.20-16.el10.noarch
Installing      : python3-cffi-1.16.0-7.el10.x86_64
Installing      : python3-cryptography-43.0.0-4.el10.x86_64
Installing      : python3-awscrt-0.27.2-1.el10_0.x86_64
Installing      : python3-docutils-0.20.1-5.el10.noarch
Running scriptlet: groff-1.23.0-10.el10.x86_64
Installing      : groff-1.23.0-10.el10.x86_64
Running scriptlet: groff-1.23.0-10.el10.x86_64
Installing      : python3-jmespath-1.0.1-8.el10.noarch
Installing      : python3-colorama-0.4.6-13.el10.noarch
Installing      : awscli2-2.27.0-1.el10_0.noarch
Running scriptlet: awscli2-2.27.0-1.el10_0.noarch
Installed products updated.

Installing:
awscli2-2.27.0-1.el10_0.noarch
python3-cffi-1.16.0-7.el10.x86_64
python3-docutils-0.20.1-5.el10.noarch
python3-prompt-toolkit-3.0.41-5.el10.noarch
python3-ruamel-yaml-clib-0.2.7-8.el10.x86_64
groff-1.23.0-10.el10.x86_64
python3-colorama-0.4.6-13.el10.noarch
python3-jmespath-1.0.1-8.el10.noarch
python3-pycparser-2.20-16.el10.noarch
python3-cryptography-43.0.0-4.el10.x86_64
python3-awscrt-0.27.2-1.el10_0.x86_64
python3-ply-3.11-25.el10.noarch
python3-ruamel-yaml-0.18.5-5.el10.noarch

Complete!
[ec2-user@ip-172-31-5-221 ~]$ aws configure
AWS Access Key ID [None]:
AWS Secret Access Key [None]:
Default region name [None]: ap-south-1
Default output format [None]: json
[ec2-user@ip-172-31-5-221 ~]$ ./backup_script.sh
upload: db_backups/full_backup_2026-04-13_06-36-53.sql to s3://amz-s3-sam-617309006396-ap-south-1-an/full_backup_2026-04-13_06-36-53.sql
[ec2-user@ip-172-31-5-221 ~]$ aws s3
usage: aws [options] s3 <subcommand> [parameters]
aws: error: too few arguments
[ec2-user@ip-172-31-5-221 ~]$ aws s3 ls
2026-04-13 06:31:07 amz-s3-sam-617309006396-ap-south-1-an
[ec2-user@ip-172-31-5-221 ~]$ aws s3 ls s3://amz-s3-sam-617309006396-ap-south-1-an
2026-04-13 06:36:55      2484816 full_backup_2026-04-13_06-36-53.sql
[ec2-user@ip-172-31-5-221 ~]$
```

Figure 4: Successful sync of `full_backup_2026-04-13_06-36-53.sql` to S3.

Step 4: Final Verification in S3 Console

A final check of the AWS Management Console confirms that the backup file is securely stored in the designated S3 bucket.

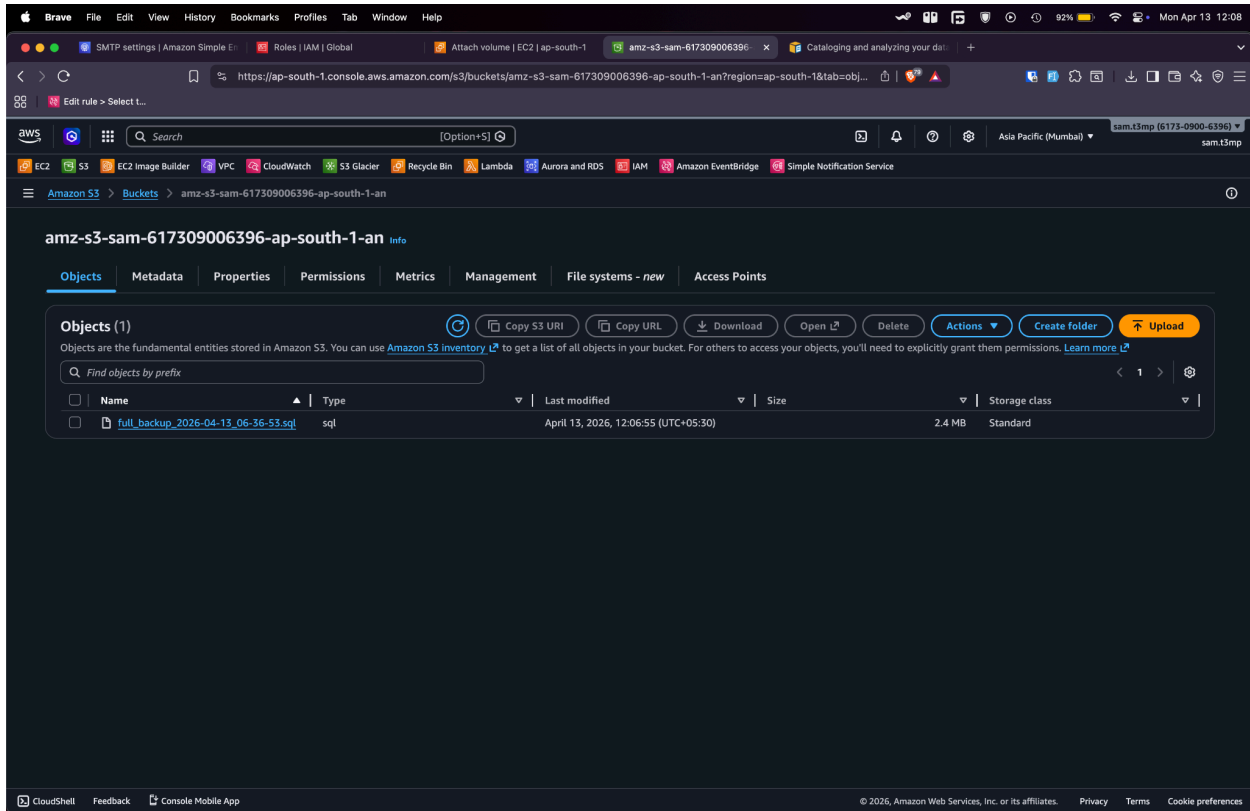


Figure 5: Verification of the backup file in the S3 bucket.